

Can Dynamic Helical Polymers be Reduced to Linear Chains of Spins? Non-Local Effects from Polymer Self-Avoidance

Keerti Chauhan, Marcus Müller, and Kostas Ch Daoulas*


Cite This: *Macromolecules* 2025, 58, 5408–5420


Read Online

ACCESS |



Metrics & More

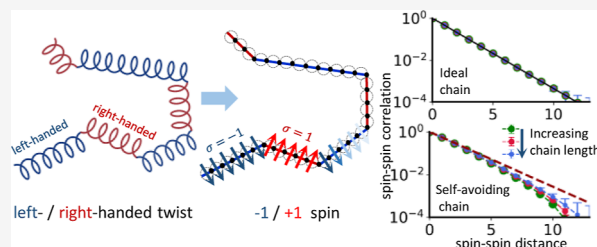


Article Recommendations



Supporting Information

ABSTRACT: Cooperative chiral order in dynamic helical polymers (DHPs) is commonly understood through their reduction to one-dimensional (1D) Ising-spin chains. In this framework, each spin, adopting the values $\sigma = -1$ and $\sigma = 1$, represents a left- and right-handed helical twist. Spins interact ferromagnetically, capturing the energetic cost of boundaries between domains of opposite chirality. All current spin models of DHPs assume that nearest-neighbor (NN) spin–spin interactions suffice to describe the system. We challenge this assumption using a minimal molecular model to investigate whether the self-avoidance of DHP chains, due to excluded volume, induces an effective, non-local coupling of spins. Our approach combines a 1D description of helicity, based on NN spin–spin interactions, with an explicit representation of DHPs as freely-jointed chains with flexible hinges. We generate both ideal and self-avoiding chain conformations through a Monte Carlo scheme. For ideal chains, the spin behavior agrees with the predictions of a NN 1D Ising model with coupling strength J . However, we find that chain self-avoidance does create effective long-range interactions, as demonstrated by spin–spin correlation functions and free-energy calculations. These interactions renormalize J and lead to finite-size effects that are inconsistent with the NN 1D Ising behavior.



1. INTRODUCTION

Covalent polymers with dynamic helical conformations represent a unique class of helical macromolecules. They have two distinctive features:^{1–3} (i) Both left- and right-handed helical states are accessible and (ii) spontaneous transitions between these two states can occur because of low helix-inversion barriers.

External chiral influences, such as chiral solvents, may break this mirror symmetry, turning these materials into permanently helical polymers with one preferred handedness. Alternatively, the handedness of dynamic helical polymers (DHPs) can be controlled by building chiral influences directly into the molecule.⁴ For example, in the sergeants-and-soldiers approach,⁵ one helical sense is promoted by “doping” a DHP with a few homochiral groups (sergeants) whereas the remaining monomers (soldiers) are achiral. In the majority-rule,⁶ the chemical composition of a DHP presents a scalemic mixture of right- and left-handed monomers. Even marginal excess of one enantiomer suffices to promote one type of handedness in the polymer helix. The responsiveness of DHPs to weak influences makes them interesting^{7–9} for applications such as chiral recognition,^{10–12} and asymmetric catalysis.¹³

Most of our understanding of the phenomenology of helical order in DHPs is derived from minimal theories^{14–19} where DHPs are reduced to one-dimensional (1D) Ising chains, with the value of each spin variable, $\sigma = \pm 1$, representing the instantaneous handedness of the respective repeat unit. Generally, it is energetically preferred for adjacent helical repeat units along a DHP chain to adopt the same helical sense (either

left- or right-handed).^{4,17} 1D Ising models account for this preference through a ferromagnetic-like coupling between spins. However, it is important to emphasize that these interactions are typically limited to nearest-neighbor (NN) interactions. The influence of local chiral effects is effectively captured by introducing an external field acting on each spin.^{4,15–17} Similar simplified models^{20,21} have also been applied to describe cooperative ordering in supramolecular DHPs.²²

The parameters of a 1D Ising model—specifically, the spin–spin coupling strength and the external field amplitude—can be determined by correlating the thermal average of the “magnetization”, $\langle S \rangle = \langle \sum_i \sigma_i \rangle$ (σ_i is the value of the i -th spin along the chain contour), with experimental measurements of optical activity in DHPs. These parameters provide valuable insights into the two fundamental molecular–level interactions: (i) the free energy cost associated with helical reversals and (ii) the strength of chiral influence acting on the repeat units.^{1,14–19}

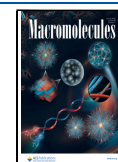
Whereas these minimal theories provide valuable insights, their basic assumptions warrant scrutiny. Polymers are inherently three-dimensional (3D) objects, where the interplay between chain connectivity and non-bonded monomer–

Received: December 22, 2024

Revised: March 31, 2025

Accepted: April 30, 2025

Published: May 18, 2025



monomer interactions gives rise to non-trivial long-range correlations. For instance, in good solvent conditions, excluded-volume interactions lead to conformations that are non-Markovian, self-avoiding, random walks. In other words, one step of the random walk depends on all the previous steps of the walk. As a result, the polymer swells, and the correlation length of bond–bond orientations becomes infinite, as manifested by the algebraic decay of the bond–bond correlation function.^{23,24}

In some DHPs helical reversals contribute to polymer flexibility^{25,26} as points where the chain can drastically change its orientation in space.⁴ Therefore, long-range correlations in bond orientation may translate into long-range correlations between points of helical reversal. In the language of 1D Ising models that would correspond to long-range correlations between spin-domain walls, or, equivalently between spin orientation (since the appearance of a domain wall requires a spin reversal). In such cases, it is unclear whether DHPs can be indeed faithfully represented by 1D Ising models with NN interactions only (referred to as NN 1D Ising models in the following). Moreover, we do not consider other sources of non-local interactions that might be present in an actual DHP, such as electrostatics.^{27,28}

Similar concerns have been raised regarding the applicability of one-dimensional (1D) models to describe the coil–helix transition,^{29,30} such as the Zimm–Bragg model³¹ and its refinements.³² These models are mathematically equivalent to NN 1D Ising models and, as a result, predict³³ that the coil–helix transition does not constitute a true phase transition. Interestingly, simulation studies provide conflicting evidence on this matter. While some support this prediction,^{30,34} others contradict it,^{29,35} using different molecular models.

Furthermore, it has been demonstrated that in dense phases of DHP, such as the crystalline or liquid-crystalline (LC) states, molecular packing effects can interfere with helicity and statistics of helical reversals. For example, experiments in crystals of poly(di-*n*-butyl silane) have found³⁶ that treatment at high pressure changes the backbone structure from helical to all trans. It has been argued^{4,37} that the close packing of aligned polyisocyanates in a LC phase reduces the population of helical reversals in order to avoid conformational kinks. These defects act as³⁷ “bad neighbors” to the local LC arrangement.

The aim of our study is to focus on isolated DHP (limit of infinite dilution) and explore to what extent generic excluded-volume effects impact the validity of their representation by NN 1D Ising models. To this end, we adapt an established minimal model^{38–40} of a freely-jointed chain (FJC) with reversible hinges. A DHP is represented by a chain of multiple links, each corresponding to a single helical turn and assigned a spin variable. The sign of the spin, $\sigma_i = \pm 1$, represents the helical sense of the turn (left- or right-handed). A standard ferromagnetic Ising coupling with strength J captures the energetic preference for adjacent turns to share the same helical sense. Sequential links with the same spin sign form a rigid segment, representing a homochiral domain. If a spin reversal occurs, the rigid segment terminates, and a freely-jointed hinge connects two adjacent domains with opposite helical senses. This approach creates a model with a strong coupling between the local helicity (spin statistics) and the overall polymer conformation. To incorporate excluded-volume effects, each link is represented by a hard sphere, preventing links from overlapping. We stochastically generate DHP conformations of

this model, using a straightforward Monte Carlo simulation scheme.

As expected, Ising spin and configurational degrees of freedom are statistically independent in the absence of excluded-volume interaction, and the behavior of the spin variables—representing the local helical sense—reproduces the phenomenology of the NN 1D Ising model. However, for self-avoiding chains with excluded volume, finite chain-length effects in the spin–spin correlations reveal the presence of an effective non-local coupling between spins. This coupling decays sufficiently rapidly with spin distance, ensuring that the system remains in the same universality class as the NN 1D Ising model. By combining excess free-energy calculations from simulations with a qualitative description of the reference free energy for ideal chains, we show that, in the long-chain limit, a DHP with excluded volume can still be described by a NN 1D Ising model but with a renormalized strength of coupling between NN spins.

2. MINIMAL MODEL

2.1. Formulating the Model. Our minimal model is a reformulation, in context of DHPs, of the well-known freely-jointed chain model with reversible hinges.^{38–40} As illustrated in Figure 1, a DHP is represented as a three-dimensional (3D)

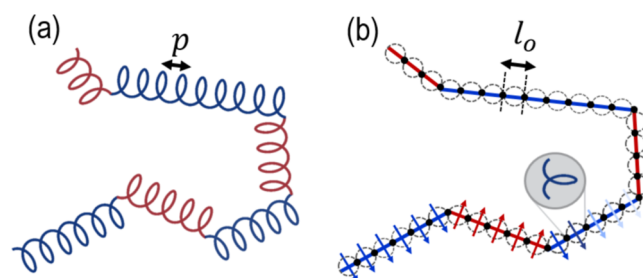


Figure 1. (a) Schematic illustration of a generic dynamic helical polymer (DHP). A DHP conformation consists of M helical turns, each with a pitch p . Individual turns can randomly switch their helical sense between right-handed (red) and left-handed (blue) states with equal probability. However, neighboring turns energetically prefer to share the same handedness, resulting in the formation of transient homochiral (right- or left-handed) domains. These domains are assumed to be stiff, and the DHP can change its direction in three-dimensional (3D) space only at points of helical reversal (domain walls). (b) Equivalent representation of the DHP conformation. The DHP is modeled as a chain with M links of length l_0 , connected by hinges (black dots). Each link i represents a single helical turn of the DHP, hence $l_0 = p$, and carries a spin variable $\sigma_i = \pm 1$, corresponding to a right-handed (+1) or left-handed (−1) helical sense. The chain behaves as freely jointed only at hinges located between links with opposing spin states. To account for excluded-volume effects, a hard sphere of diameter $2R = l_0$ is placed at the center of mass of each link. Further details are provided in the main text.

chain consisting of M connected links (or bonds) of fixed length l_0 . Each link corresponds to a group of monomeric units forming one helical turn in the actual polymer, such that l_0 equals the helical pitch p . To capture the helical sense of the underlying turn, each link is assigned a spin variable $\sigma_i = \pm 1$, where i denotes the ranking number (position) of the link along the chain contour ($i = 1, \dots, M$). The cooperative behavior of DHPs—specifically, the energetic preference for successive helical turns to share the same handedness—is modeled by coupling adjacent spins in the same way as in the NN 1D Ising model of DHPs. This coupling corresponds to the Hamiltonian

$$\beta H_{\text{Ising}} = -J_0 \sum_{i=1}^{M-1} \sigma_i \sigma_{i+1} \quad (1)$$

The positive coefficient J_0 controls the coupling strength. Notably, βH_{Ising} does not impose any constraints on the states of the first and last spins in the chain. This setup is known as the free-boundary condition. To keep the model simple, J_0 is assumed to be independent on the position of the NN spins in the chain. This approach, for example, neglects that side chains of underlying DHP monomers may interact with each other more loosely at chain ends (reduced cooperativity), making it easier for the DHP backbone rotations to occur there than in the middle of the DHP.²⁶ Refinements considering a position-dependent J_0 are possible, introducing some similarity to the “tailed-wormlike chain” model.⁴¹ Throughout this work, all interactions will be expressed in units of the thermal energy $k_B T$ ($\beta = 1/k_B T$), and all lengths in units of the bond length l_0 .

In a DHP, a group of successive helical turns with the same handedness forms a homochiral domain. These homochiral domains are assumed to be stiff. In our minimal model, this assumption implies that consecutive links with identical spin values are strictly aligned (see Figure 1). Consequently, a domain consisting of m same-signed spins corresponds to a rigid segment of length ml_0 . Conversely, helical turns with opposite handedness are assumed to orient randomly with respect to each other in space. In our minimal model, this scenario is realized by introducing fully flexible hinges between bonds whose spin variables have opposite signs. This distinctive chain flexibility is formalized via the Hamiltonian

$$\beta H_{\text{bend}} = \sum_{i=1}^{M-1} U_{\text{bend}}(\sigma_i, \sigma_{i+1}, \mathbf{t}_i, \mathbf{t}_{i+1}) \quad (2)$$

where $\mathbf{t}_i$ is a unit vector oriented along the i -th bond. The spin-dependent angular potential U_{bend} is defined as

$$U_{\text{bend}}(\sigma_i, \sigma_{i+1}, \mathbf{t}_i, \mathbf{t}_{i+1}) = \begin{cases} 0 & \text{if } \sigma_i \neq \sigma_{i+1}, \text{ or } \sigma_i = \sigma_{i+1} \text{ and } \mathbf{t}_i \uparrow \uparrow \mathbf{t}_{i+1} \\ \infty & \text{if } \sigma_i = \sigma_{i+1} \text{ and } \mathbf{t}_i \nparallel \mathbf{t}_{i+1} \end{cases} \quad (3)$$

Finally, we assume that each bond is surrounded by a spherical hard excluded volume with a diameter $d = 2R$, equal to its length l_0 . This assumption is reasonable since $l_0 \sim p$, and experimental studies typically report that the thickness d of DHP chains is only moderately larger than the helical pitch p . For example, certain polysilylenes²⁶ adopt a 7/3 helix with a translation of $h \simeq 0.196$ nm per monomer, which corresponds to $p \simeq 0.46$ nm in our convention. The same polysilylenes exhibit a thickness of approximately $d \simeq 1.8$ nm.

The excluded-volume interactions are described in our model by the following Hamiltonian

$$\beta H_{\text{ex}} = \sum_{i=1}^{N-2} \sum_{j=i+2}^N U_{\text{ex}}(|\mathbf{r}_{ij}|) \quad (4)$$

Here $\mathbf{r}_{ij}$ denotes the distance vector connecting the centers-of-mass (COM) of the i -th and j -th bonds, and $U_{\text{ex}}(|\mathbf{r}_{ij}|) = +\infty$ if $|\mathbf{r}_{ij}| < 2R$ and $U_{\text{ex}}(|\mathbf{r}_{ij}|) = 0$ otherwise. There are no excluded-volume interactions between nearest neighbors, i.e. bonds i and $j = i + 1$. This feature of the model is reflected in the limits of the summations in eq 4.

To demonstrate that, for ideal chains—where excluded-volume interactions βH_{ex} are deactivated—our minimal model is equivalent to the model in the version of Glatting et al.,³⁹ we reproduce their single-chain partition function

$$Z(T) = Z_0(T) \int d\mathbf{b}_1 \frac{\delta(|\mathbf{b}_1| - l_0)}{4\pi l_0^2} \prod_{i=2}^M \left\{ \int d\mathbf{b}_i \frac{\delta(|\mathbf{b}_i| - l_0)}{4\pi l_0^2} \times \left[1 + \frac{4\pi c}{\sin(\theta_i)} \delta(\theta_i - \theta_{i-1}) \delta(\phi_i - \phi_{i-1}) \right] \right\} \quad (5)$$

Here $\mathbf{b}_i$ represents the vector of the i -th bond, with its length constrained to l_0 by the corresponding δ -function. The angular coordinates of this bond are given by θ_i and ϕ_i . The factor c denotes the ratio of probabilities of a rigidly aligned hinge and a fully flexible one

$$c = \frac{e^J}{4\pi e^{-J_0}} \equiv e^{2J} \text{ with } J \equiv J_0 - \frac{\ln 4\pi}{2} \quad (6)$$

i.e. $2J$ is the free-energy cost of breaking a homochiral domain, comprised of the energy increase $2J_0$ and the configurational entropy gain $\ln 4\pi$. Therefore, J is often denoted as cooperativity parameter. The prefactor $Z_0(T)$ depends on temperature T and stems from the integration over momenta.³⁹ It is irrelevant when calculating conformational averages. Noh and Benetatos⁴⁰ explicitly connected c to an activation energy via an Arrhenius expression. Glatting et al.³⁹ directly scanned the parameter space of c .

Glatting et al.³⁹ provide a useful analytical expression for the mean squared end-to-end distance $\langle R_e^2 \rangle_{\text{id}}$ of an ideal chain

$$\langle R_e^2 \rangle_{\text{id}} = \frac{Ml_0^2}{[1+c]^{M-1}} \sum_{N=1}^M c^{M-N} \binom{M-1}{N-1} \frac{2M-(N-1)}{N+1} \quad (7)$$

The round brackets represent the binomial coefficient. An approximate result, valid when many freely-jointed hinges are present, was also derived³⁹

$$\langle R_e^2 \rangle_0 = Ml_0^2 [1+2c] \quad (8)$$

Another useful property is the average number of freely-jointed hinges in an ideal chain, or equivalently, the average number of domain walls $\langle N_d \rangle$, which is given by an exact expression derived from the 1D NN Ising model⁴²

$$\langle N_d \rangle = (M-1) \frac{e^{-J}}{e^{+J} + e^{-J}} \quad (9)$$

The model introduced here does not explicitly capture the helical shape of the DHP. However, this minimalistic approach offers significant advantages for our purposes, as in the ideal-chain limit it is equivalent to the NN 1D Ising models of DHPs.^{14–19} In this limit, only the interactions βH_{Ising} and βH_{bend} are present. As a result, there is no statistical correlation between the polymer's spatial arrangement and the spin states (i.e. the local helical sense). By adding excluded-volume interactions, βH_{ex} , into this reference system, we can assess whether chain self-avoidance influences the spin states, leading to deviations from the NN 1D Ising behavior.

2.2. Sampling Method. Previous studies^{39,40} have shown that for ideal chains, this model can be analyzed analytically. However, the inclusion of excluded-volume interactions complicates the problem. To analyze the statistics of spin variables and chain conformations in this case, we adopt a simple sampling approach, generating conformations distributed according to the Boltzmann weight

$$W_B = \exp[-\beta(H_{\text{Ising}} + H_{\text{bend}} + H_{\text{ex}})] \quad (10)$$

To generate a chain conformation, we begin by linearly adding M bonds. The two angular coordinates, θ_1 and ϕ_1 , along with the spin σ_1 of the first bond, are selected randomly. Next, we choose the spin of the second bond as $\sigma_2 = -\sigma_1$ with probability $r = \frac{4\pi e^{-J_0}}{e^{J_0} + 4\pi e^{-J_0}} = \frac{e^{-J}}{e^J + e^{-J}}$, and $\sigma_2 = \sigma_1$ with probability $1 - r$. If $\sigma_2 = -\sigma_1$, the angular coordinates θ_2 and ϕ_2 are randomly chosen on a unit sphere with probability $1/4\pi$. If $\sigma_2 = \sigma_1$, the second bond is added as a straight extension of the first, meaning $\theta_2 = \theta_1$ and $\phi_2 = \phi_1$. This procedure is similarly repeated for the remaining bonds: the spin σ_i and the angular coordinates θ_i and ϕ_i of the i -th bond are proposed based on the previous bond's values, σ_{i-1} , θ_{i-1} , and ϕ_{i-1} . Once the conformation of the entire chain is generated, we check for excluded-volume overlaps between segments. If the distance between the center-of-mass (COM) of any two segments (excluding nearest neighbors) is smaller than $2R$, the conformation is discarded. Otherwise, it is added to the sample.

For ideal chains, this method is attrition free. By running the scheme in parallel on 32 cores, we were able to generate samples containing millions of conformations with modest computational effort. For example, 10 days of sampling were sufficient for the most demanding cases considered in this study. For ideal chains and a typical moderate polymer length (e.g. $M = 200$) the sample contained 160×10^6 conformations. For longer polymers (e.g. $M = 1000$), we typically reduced the size to 1.5×10^6 conformations.

For non-ideal chains, however, the average fraction $\langle W \rangle$ of generated conformations that survives the excluded-volume screening decreases exponentially with M .^{43,44} Nevertheless, even in this case, our sampling method was able to generate large samples of conformations for chains up to about hundred times longer than the persistence length (at each considered J). For example, for $J = 0.5$ and $M = 200$ we typically analyzed 25×10^6 conformations. Large sample sizes are necessary to obtain reliable statistics, particularly for spin–spin correlation functions.

This direct excluded-volume screening approach enables a straightforward calculation of the free-energy difference, $\beta\Delta F$, between self-avoiding and ideal chains in our model. The excess free energy is given by⁴⁴

$$\beta\Delta F = -\ln[\langle W \rangle] \quad (11)$$

3. CONFORMATIONAL PROPERTIES

We begin by quantifying standard conformational properties, which will provide a foundation for analyzing and interpreting the behavior of the spin variables (i.e., the local helical sense) in the subsequent Section 4.

For ideal chains, we compare the dependence of $\langle R_e^2 \rangle_{\text{id}}$ on the number of links M in the simulations with the theoretical prediction³⁹ given in eq 7. Figure 2 presents results for two representative values of the cooperativity parameter: $J = 0.5$

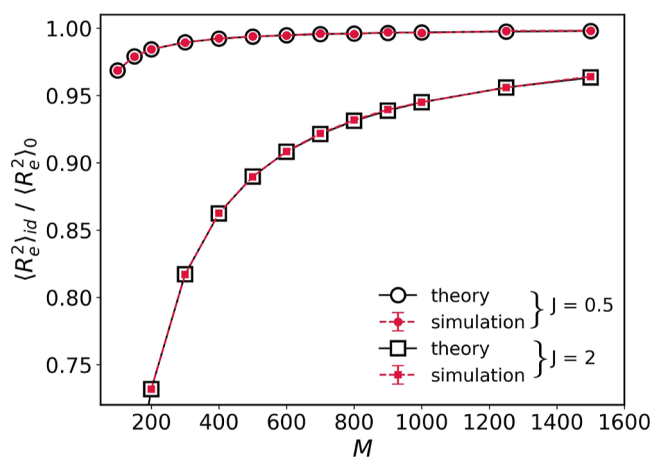


Figure 2. Mean squared end-to-end distance, $\langle R_e^2 \rangle_{\text{id}}$, of ideal polymer chains, scaled by the large- M limit $\langle R_e^2 \rangle_0$ predicted by theory³⁹ (see eq 8). Two representative coupling strengths, $J = 0.5$ (corresponding to $c \approx 2.718$) and $J = 2$ (corresponding to $c \approx 54.598$), are considered. The results of the simulations (red solid symbols) are compared with theoretical predictions³⁹ (black open symbols).

(corresponding to $c \approx 2.718$) and $J = 2$ (corresponding to $c \approx 54.598$). Chains with lengths up to $M = 1500$ segments are considered, and $\langle R_e^2 \rangle_{\text{id}}$ is normalized by the approximate result $\langle R_e^2 \rangle_0$ (see eq 8), which is expected to be accurate for $M \gg 1$, where the number of freely-jointed hinges is large. The simulation data and theoretical predictions coincide (within the small error bars). For $J = 0.5$, the normalized $\langle R_e^2 \rangle_{\text{id}}$ approaches unity, which is consistent with expectations. From eq 9, one can straightforwardly verify that for $J = 0.5$ and $M \sim 10^3$, the average number of freely-jointed hinges (domain walls) is $\langle N_d \rangle \sim 10^2$. In contrast, for the same chain lengths but with a substantially stronger coupling of $J = 2$, the number of reversals is reduced to $\langle N_d \rangle \sim 10$, and the normalized $\langle R_e^2 \rangle_{\text{id}}$ is clearly below unity.

The excluded-volume interactions affect the local chain stiffness, for example by canceling configurations where close bonds overlap. Therefore, the local stiffness of our self-avoiding chains cannot be quantified via Kuhn segments calculated in the ideal chain limit. It must be independently characterized by an effective Kuhn segment $l_{k,\text{ex}}$ that takes into account local effects of excluded volume, to ensure that during the analysis of spin–spin correlations in polymers with excluded volume for different J (see Section 4), we compare chains with the same number of effective Kuhn segments, $N_k = Ml_0/l_{k,\text{ex}}(J)$.

Determining an effective Kuhn segment, equivalently an effective persistence length, of a self-avoiding polymer is not straightforward.⁴⁵ We follow a pragmatic approach⁴⁶ and calculate the bond–bond correlation function $C_b(s) = \langle \mathbf{t}_i \cdot \mathbf{t}_{i+s} \rangle$, where s is the difference of the ranking numbers of two bonds along the chain contour (chemical distance). For a few representative chain lengths, Figure 3 presents the $C_b(s)$ for $J = 0.5$ and $J = 2$. In contrast to the exponential decay of bond–bond correlations in ideal chains,^{47,48} self-avoiding polymers exhibit an algebraic decay $C_b(s) \sim s^{-2(1-\nu)}$ with $\nu \approx 0.588$ for $s \gg 1$.^{23,24} However, we only require an approximate estimate of the local length scale at which the self-avoiding polymers become flexible. Therefore, we define $l_{k,\text{ex}}$ by fitting $C_b(s)$ with an exponential decay $\exp(-s/l_{k,\text{ex}}(J))$ for small s , where $0.13 \leq C_b(s) \leq 1$. In this range, the bond–bond correlation function is

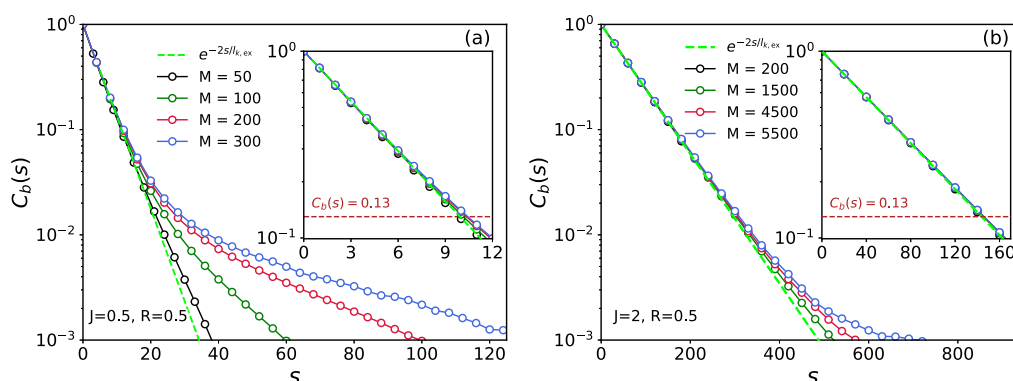


Figure 3. Linear-log plots of the bond–bond correlation function $C_b(s)$, where s denotes the contour distance between two bonds along the polymer chain. Panels (a,b) depict results for polymers with $J = 0.5$ and $J = 2$, respectively. For each value of J , $C_b(s)$ is calculated for several representative chain lengths M . The green dashed line represents a fit to an exponential-decay function, applied over the range $0.13 \leq C_b(s) \leq 1$. This small- s region is highlighted in the insets, together with the exponential fit.

reasonably well-approximated by an exponential decay, although some effects of finite chain length are evident—most notably for the more flexible chains, $J = 0.5$; see insets of Figure 3a,b. This small- s behavior has an analogy in ideal chains, where $C_b(s) \simeq 0.13$ corresponds to an s that roughly matches the usual Kuhn segment l_k of an ideal chain.

We emphasize that $l_{k,ex}$ characterizes only the local chain stiffness and does not imply that the large-scale behavior of a self-avoiding coil (above the scale of a thermal blob) can be reduced to ideal Kuhn chains. We obtain $l_{k,ex}(0.5) \simeq 9.74$, $l_{k,ex}(1) \simeq 21.94$, and $l_{k,ex}(2) \simeq 141.1$. Additional data provided in the Supporting Information demonstrate that $l_{k,ex}(J)$ in chains with excluded volume is systematically larger than the Kuhn segment of ideal chains at the same J and M (based on exponential fits of $C_b(s)$ for ideal chains).

On physical grounds, the diameter of the spherical excluded volume was chosen to be small, $d = l_0$. Therefore, it is essential to quantify the swelling behavior of the polymer chains for the degrees of polymerization considered in our study. The main panel of Figure 4 presents the squared expansion factor α^2 , defined as the ratio of the mean squared end-to-end distance for chains with excluded volume to that of ideal chains:⁴⁷ $\alpha^2 = \langle R_e^2 \rangle_{ex} / \langle R_e^2 \rangle_{id}$. A log–linear plot of α^2 is presented for several representative values of J , as a function of the scaled excluded-volume parameter $\tilde{z}$, which is defined as⁴⁹

$$\tilde{z} = (\nu/l_0^3)K(N_k)\sqrt{N_k} \frac{l_0}{l_{k,ex}(J)} \quad (12)$$

Here, ν is the excluded volume of the bonds, that is, a “microscopic” parameter that does not depend on J . Up to numerical prefactors independent of N_k and J , $\tilde{z}$ serves as the scaled excluded-volume parameter.⁴⁹ Since $K(N_k) \rightarrow 4/3$ for $N_k \rightarrow \infty$, $\tilde{z}$ is proportional to the Fixman parameter in this limit. The function $K(N_k)$ in eq 12 has been detailed by Shimada and Yamakawa.⁵⁰ For reference, this function is reproduced in the Supporting Information.

Figure 4 illustrates that the polymer chains swell with increasing $\tilde{z}$, indicating the impact of excluded volume. The data are consistent with the expectation^{51,52} that α^2 is a universal function of $\tilde{z}$. However, the chains in our study are still far from reaching the asymptotic scaling regime, $\alpha^2 \sim \tilde{z}^{2(2\nu-1)}$ with $\nu \simeq 0.588$, as predicted by renormalization-group theories⁵³ and verified in simulations⁵⁴ for $\tilde{z} \rightarrow \infty$. This is further illustrated in the inset of Figure 4.

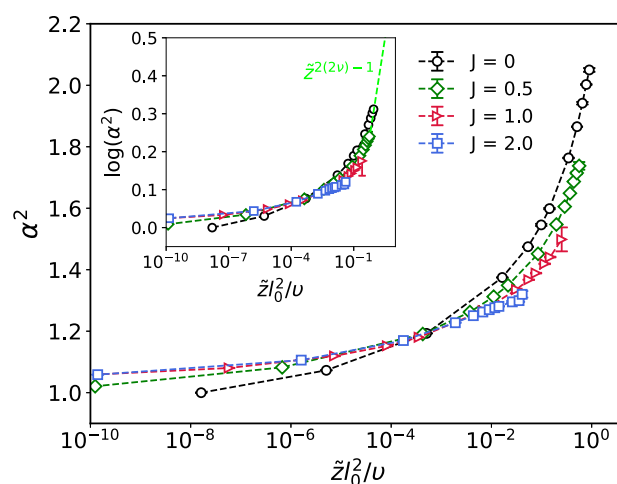


Figure 4. Main panel: log–linear plot of the squared expansion factor α^2 as a function of $\tilde{z}l_0^2/\nu$ for several values of the coupling strength J . The scaled excluded-volume parameter $\tilde{z}$ is defined in eq 12. ν and l_0 are the excluded volume and the length of the bonds, respectively. They do not depend on J and M , and are, therefore, immaterial for the scaling plot. Inset: log–linear plot of $\log \alpha^2$ as a function of $\tilde{z}l_0^2/\nu$, providing a comparison with the expected scaling behavior (green dashed line) in the limit $\tilde{z}l_0^2/\nu \rightarrow \infty$.

4. CORRELATIONS OF SPIN VARIABLES (LOCAL HELICAL SENSE)

To demonstrate that, for ideal chains, the behavior of spin variables (local helical sense) reproduces the one of a NN 1D Ising model, we calculate the spin–spin correlation function $C(s) = \langle \sigma_i \sigma_{i+s} \rangle$. Analogous to the bond–bond correlation function, s stands for the chemical distance between two spins along the chain contour while the angular brackets denote an average over configurations and spin positions i . In a NN 1D Ising model with free boundary conditions the $C(s)$ is independent^{55,56} from the particular position of two spins in the Ising chain (i.e. the correlation depends only on spin distance s) and is given by

$$C(s) = \exp[-s/\xi], \quad \text{with } \xi = 1/|\ln(\tanh(J))| \quad (13)$$

Figure 5 presents a linear-log plot of $C(s)$ for ideal chains as a function of s normalized by the correlation length ξ , for two characteristic cooperativity parameters, $J = 0.5$ and $J = 2$, and two

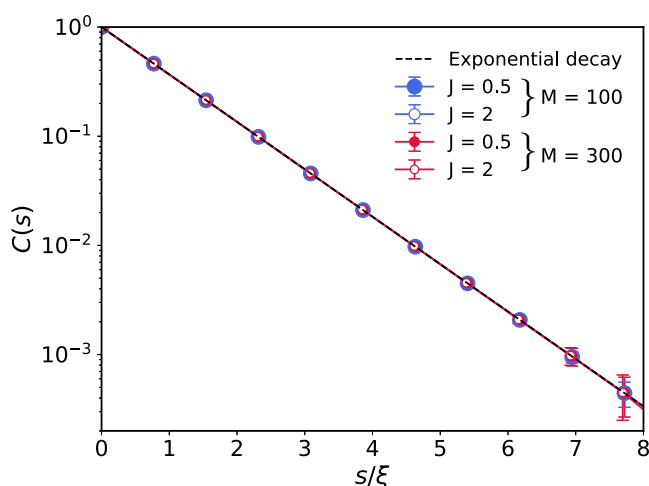


Figure 5. Correlation function, $C(s)$, of spin variables (local helical sense) in ideal chains as a function of the normalized distance s/ξ . Results are shown for two combinations of coupling strength, $J = 0.5$ (solid circles) and $J = 2$ (open circles), and chain lengths, $M = 100$ (blue) and $M = 300$ (red). The black dashed line represents the prediction of the NN 1D Ising model, eq 13.

polymer lengths, $M = 100$ and 300 . For all cases presented, we can accurately determine the correlation function for $C(s) \geq 10^{-3}$. Within the error bars, the data perfectly agree with the master curve of the NN 1D Ising, given by eq 13.

We now turn our attention to self-avoiding chains. Theories of long-range interactions in 1D Ising models typically introduce a non-local pair-coupling coefficient, $J(s)$, that is a function of the distance s between interacting spins. The phase behavior depends on how the spin–spin coupling $J(s)$ decays with s .^{57–59} For algebraic decays, $J(s) \sim s^{-\lambda}$, long-range order appears at temperatures below a finite critical temperature, $T_c > 0$, if $1 < \lambda \leq 2$.^{57–59} The behavior for $\lambda = 2$ is of special fundamental interest as a 1D analog^{60,61} of the Kosterlitz–Thouless transition.⁶² If $\lambda > 2$ there is no long-range order at any finite temperature,^{58,63} and the system belongs to the same universality class as the NN 1D Ising model.⁶⁴

The introduction of excluded-volume interactions in DHPs, however, does not induce long-range order of the local helical sense (spins), at least for the (realistic) magnitudes of R considered in this study. To demonstrate this, we examine the Binder cumulant, U_M ,⁶⁵ of the order parameter S ¹⁵

$$U_M = \frac{3}{2} \left(1 - \frac{1}{3} \frac{\langle S^4 \rangle}{\langle S^2 \rangle^2} \right) \quad (14)$$

$$\text{with } S = \sum_{i=1}^M \sigma_i \quad (15)$$

Recall that, according to the analogy between spin order and local helical sense, the thermal average of the “magnetization” $\langle S \rangle$ quantifies the average excess of one helical sense over the other. If excluded-volume interactions had induced a spontaneous breaking of mirror symmetry, the phase diagram of our DHP in the $J - \langle S \rangle$ plane would exhibit two symmetric branches of magnetization—one positive and one negative—at sufficiently large J .

The main panel of Figure 6 presents the dependence of U_M on J , calculated for several system sizes (chain lengths M). The plots of U_M for different M exhibit no tendency to intersect,

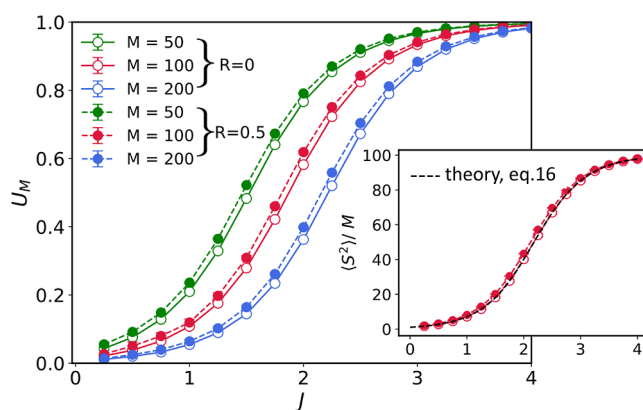


Figure 6. Main panel presents the Binder cumulant, U_M , as a function of coupling strength, J , for three chain lengths, M . Results are presented for simulations without (open circles) and with (solid circles) excluded-volume interactions. Notably, data sets for different chain size, M , do not cross. The inset presents the mean square of “magnetization” (excess helicity) $\langle S^2 \rangle / M$ as a function of J , for one characteristic chain length $M = 100$, calculated for ideal chains (open circles) and chains with excluded volume (solid circles). The black dashed line is the prediction for the NN 1D Ising model, eq 16.

demonstrating the absence of a phase transition.^{66,67} Moreover, the dependence of U_M on M for chains with excluded-volume interactions resembles and qualitatively matches the behavior of U_M calculated for the same chain lengths in ideal polymers ($R = 0$).

The strength of fluctuations of helical sense in a DHP can be also quantified by calculating $\langle S^2 \rangle / M$ as a function of J . For NN 1D Ising models with free boundary conditions⁵⁵

$$\frac{\langle S^2 \rangle}{M} = \frac{1 + \tanh(J)}{1 - \tanh(J)} - \frac{2 \tanh(J)}{M} \frac{1 - \tanh(J)^M}{(1 - \tanh(J))^2} \quad (16)$$

In the thermodynamic limit, $M \rightarrow +\infty$, eq 16 reduces to $\langle S^2 \rangle / M = \exp(2J)$.

The inset of Figure 6 presents $\langle S^2 \rangle / M$ as a function of J , for one representative chain length $M = 100$, calculated in simulations of ideal chains and chains with excluded volume. For ideal chains, the plot agrees (within error bars) with eq 16. Qualitatively, the fluctuations in self-avoiding chains follow the same trend. Overall, the model predicts that the fluctuations in helical order increase as the cooperativity becomes larger. This behavior can be rationalized considering that in DHP with strong cooperativity, a chain is comprised of fewer domains, giving rise to an increase in fluctuations of S . This prediction applies to equilibrium fluctuations only, where the system is allowed sufficient time to sample the entire configurational space.

The absence of a phase transition can still be consistent with couplings of distant spins, that decay faster than $\sim 1/s^2$ yet nevertheless influence the disordered state. To probe such non-local couplings, we analyze the spin–spin correlation function, $C(s)$.

To elucidate the effect of excluded-volume interactions between distant parts of the chain on spin–spin correlations, it is first helpful to perform simulations where excluded-volume interactions are activated only upto n -nearest neighbors (n -NN). We start from 3-NN (recall that excluded-volume interactions between NN segments are never included). This is the next-nearest neighbors (NNN) scenario, where overlaps are checked only for segments that are separated by a chemical distance $s = 2$.

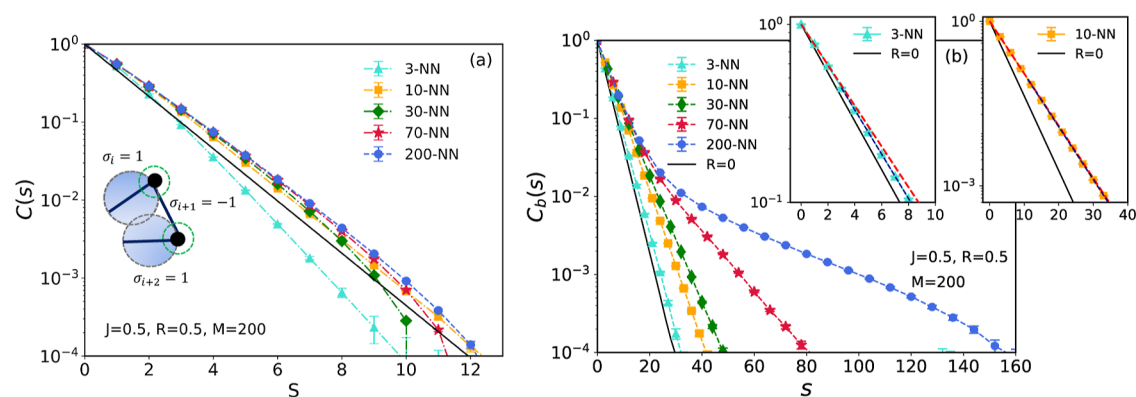


Figure 7. Panel (a) presents correlation functions of spin variables $C(s)$, as a function of the contour distance s between two spins, obtained in simulations where excluded-volume interactions are activated only upto n -nearest neighbors (n -NN). Chains with $J = 0.5$ and $M = 200$ are considered; therefore the plot for 200-NN refers to the case where all excluded-volume interactions are present. The inset includes a schematic representation of an overlapping (forbidden) triplet configuration, $\sigma_i = 1$, $\sigma_{i+1} = -1$, $\sigma_{i+2} = 1$, for a chain with only NNN excluded-volume interactions. Due to the alternating spin values, the bonds are free to rotate around the two connecting hinges (marked by green circles). Panel (b) presents the bond–bond correlation functions $C_b(s)$ calculated in simulations shown in panel (a). The two insets highlight the behavior of $C_b(s)$ for two small representative values 3-NN and 10-NN. The exponential decay of an ideal chain is recovered for 3-NN and 10-NN when $s \geq 3$ and $s \geq 10$, respectively. In the two insets this decay is indicated by black dashed lines. The red dashed lines indicate the change of slope for $s < n$ -NN where excluded-volume interactions are active. The black solid line marks the $C_b(s)$ of the ideal chain.

Next, for 4-NN overlaps are checked only for segments that are separated by a chemical distance $s = 2$ or $s = 3$. We note that the large-scale chain configurations remain Gaussian in the limit of finite n but $M \rightarrow \infty$. The n -NN is further systematically increased until reaching the limit where excluded-volume interactions are present between all segments, i.e. the original model is recovered, see eq 4. We indicate this limit of full blown excluded-volume interactions by M -NN.

Figure 7a presents the $C(s)$ calculated in simulations of chains with $J = 0.5$ and $M = 200$, for representative choices of n -NN. For 3-NN the $C(s)$ is clearly below the correlation function of the NN 1D Ising model already for $s > 2$. Involving excluded-volume interactions between a handful of segments—for example, see the $C(s)$ for 10-NN simulations—increases correlations. This effective increase of spin coupling constant is linked to local packing of monomers only. Because repulsions between only a few sequential monomers are involved, the chain remains ideal on large scales, as is illustrated by the corresponding bond–bond correlation functions $C_b(s)$ in Figure 7b. The two insets of Figure 7b demonstrate that the exponential decay of an ideal chain is recovered for 3-NN and 10-NN when $s \geq 3$ and $s \geq 10$ respectively.

When the excluded-volume interactions begin to involve larger parts of the chain, e.g. 30-NN, the chain configurations swell. This swelling of the polymer becomes evident in the plots of $C_b(s)$, as deviations from the exponential large- s behavior. In the limit $1 \ll n$ -NN $\ll M$, we would expect the excluded-volume power law in the regime $1 \ll s \ll n$ -NN but this limit is not reached in our simulations.

Excluded-volume effects are also detectable in the spin–spin correlation, $C(s)$, that becomes non-exponential. As the range of n -NN interactions becomes comparable to M the chain swells further, and the spin–spin correlations increase a bit in the tail but the overall shape of $C(s)$ remains non-exponential.

Explaining the changes of $C(s)$ at intermediate values of n -NN in simple terms is challenging. This requires the enumeration of a large number of polymer conformations and understanding how the cancellation of conformations by chain overlaps, in this set, affects spin statistics. In any case, for all n -NN, the spin statistics adjusts to ensure self-avoidance and swelling of the

polymers. Figure 7b clearly demonstrates that the bond–bond correlations increase systematically as more segments are added to the excluded-volume interactions.

An enumeration of conformations is straightforward only in the case of 3-NN and is provided in the Supporting Information. This analysis demonstrates that the significant drop of the $C(s)$ below the correlation function of the NN 1D Ising model stems from a weak, effective antiferromagnetic-like coupling between NNN spins. This coupling emerges by the elimination of a subset of chain conformations due to NNN segment overlaps, specifically those corresponding to two distinct spin triplets: $\sigma_i = 1$, $\sigma_{i+1} = -1$, $\sigma_{i+2} = 1$ and $\sigma_i = -1$, $\sigma_{i+1} = 1$, $\sigma_{i+2} = -1$. The inset of Figure 7a illustrates such an overlapping triplet configuration.

The analogous reduction of the correlation functions, compared to the reference NN Ising model, is consistent with the analytical results of Stephenson⁶⁸ for 1D Ising models with NN ferromagnetic and NNN antiferromagnetic couplings.

We now can explore systematically the behavior of spin–spin correlations as a function of polymer length in simulations where all excluded-volume interactions are present in the chain. Figure 8 presents the $C(s)$ for $J = 0.5$, and three chain lengths, $M = 50$, 100, and 200 (solid circles). The correlation function of the NN 1D Ising model (ideal chain) with $J = 0.5$ is also presented (solid black line). We observe that the spin–spin correlations in the region of small s seem to systematically increase as chains become longer. Furthermore, the crossover of the $\ln C(s)$ to the nonlinear regime shifts to larger contour distances s for longer chains. We attribute the two effects of increasing chain length—the growth of spin–spin correlations and shift of the crossover—to effective, excluded-volume-mediated couplings between distant spins.

For infinite chains, we hypothesize that the correlation functions will eventually converge to a linear dependence of $\ln C(s)$ consistent with a NN 1D Ising model but with an effective, renormalized coupling J' . This linear behavior of $\ln C(s)$ is illustrated in Figure 8 by the red dashed line. The effective, renormalized coupling J' can be estimated as $J' = \text{atanh}[C(s = 1)]$. Further evidence is provided in Section 5.

Another evidence for effective long-range interactions can be obtained from the position-dependent spin–spin correlation

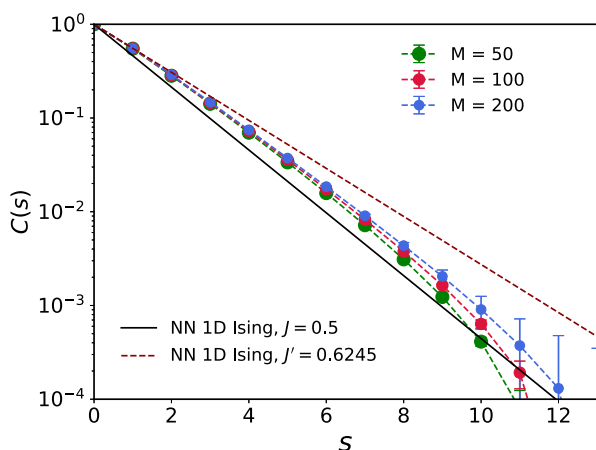


Figure 8. Correlation function of spin variables $C(s)$, as a function of the distance s between two spins along the chain contour. $C(s)$ is calculated for chains with excluded-volume interactions and coupling strength $J = 0.5$, for three chain lengths M (solid circles). The black solid line presents the $C(s)$ in the NN 1D Ising model with $J = 0.5$, whereas the red dashed line is the $C(s)$ of the NN 1D Ising model with renormalized J' (details are provided in the main text).

function $C(i, s) = \langle \sigma_i \sigma_{i+s} \rangle$. This quantity explicitly accounts for the different positions of the reference spin i along the chain contour, i.e., here the angular brackets stand only for a configurational average. Figure 9 presents (open circles) the

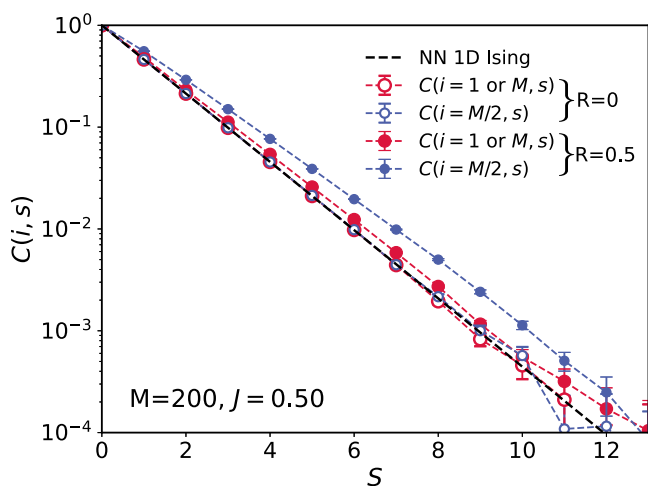


Figure 9. Position-dependent spin–spin correlation function $C(i, s)$ plotted as a function of the contour distance s between two spins, when the reference spin i is located on one of the two ends of the chain ($i = 0$ or $i = M$, red circles) and in the middle of the chain ($i = M/2$, blue circles). The plots with open and solid circles have been calculated, respectively, for an ideal chain and a chain with excluded volume. In both cases, $J = 0.5$ and $M = 200$. The black solid line presents the prediction of the NN 1D Ising model for the spin–spin correlation function which depends only on the distance of two spins but not their position in the chain, cf. eq 13.

$C(i, s)$ calculated for an ideal chain with $J = 0.5$ and $M = 200$ when i is located on one of the two ends of the chain ($i = 1$ or $i = M$, red circles) and in the middle of the chain ($i = M/2$, blue circles). As expected for a NN 1D Ising behavior, the curves are identical within error bars and, furthermore, match eq 13. However, the situation changes for a chain with excluded volume. The corresponding plots are presented in Figure 9 with

solid circles, demonstrating that the correlations for the middle spin (blue circles) are clearly stronger than for the end spins (red circles). This effect stems from the truncation of some long-range interactions for the end spins. Therefore, these spins fluctuate stronger and the helical order at the ends of the polymer is less coherent.

Figure 10a,b present the spin–spin correlation function for the two stronger couplings $J = 1$ and $J = 2$, respectively. The three chain lengths, M , indicated in the key, are chosen to match the number of Kuhn segments, N_k , of DHPs with $J = 0.5$, full excluded-volume interactions, and $M = 50, 100$, and 200 .

Additionally, Figure 10a and b report $C(s)$ from simulations with only NNN excluded-volume interactions (solid triangles), which nearly overlap with the prediction of the NN 1D Ising model (solid black line) and therefore serve as the baseline for the simulations with full excluded volume. This behavior contrasts with the case of $J = 0.5$ and can be explained simply: for large J , the number, N_d , of domain walls (or “kinks”, $\sigma_i \sigma_{i+1} = -1$) per chain is small, and backfolding of the chains is rare. As a result, NNN interactions have only a slight effect on the distribution of conformations, and the spin statistics closely follow that of the ideal chain.

In Figure 10a,b, the correlation functions in simulations with full excluded-volume interactions clearly deviate from the baseline of the NN 1D Ising model. For $J = 1$, the plots of $C(s)$ show a considerably weaker dependence on chain length compared to the $J = 0.5$ data. Nevertheless, for larger s , the correlation functions still do not follow the exponential behavior expected for the NN 1D Ising model. This is evident from the deviation of $\ln C(s)$ from the expected linear behavior of the NN 1D Ising model with renormalized coupling $J' = \text{atanh}[C(s = 1)]$ (red dashed line). In the region where statistical errors allow for accurate determination of $C(s)$ for $J = 2$, we observe only a minimal chain-length dependence. The plots closely match the correlation function of the NN 1D Ising model with effective, renormalized coupling $J' = \text{atanh}[C(s = 1)]$ (red dashed line). Furthermore, as the spin–spin coupling increases, the relative difference $\delta J/J = (J' - J)/J$ between the bare J and the renormalized J' decreases. For instance, for $J = 2$ we have $\delta J/J \approx 0.03$, as opposed to $\delta J/J \approx 0.09$ for $J = 1$ and $\delta J/J \approx 0.25$ for $J = 0.5$.

5. INSIGHTS FROM THE FREE ENERGY

Phenomenologically, excluded-volume interactions cause the DHP to swell (see Figure 4), effectively reducing its self-density. This swelling is accompanied by an increase in the length of homochiral domains, reflected in the enhancement of spin–spin correlations. This interplay between polymer expansion and the alignment of local helical senses causes the increase from J to the effective, renormalized J' .

We propose that the excluded-volume effects can be effectively analyzed by considering DHPs at an even coarser level, modeled as freely-jointed chains. In this framework, the number of links corresponds to the average number, $\langle N_d \rangle + 1$, of homochiral domains (cf. Figure 1). The free energy of such a chain can be expressed as

$$\beta F = \beta \Delta F + \beta F_{\text{id}} \quad (17)$$

where βF_{id} stands for the free energy of an ideal chain, which is equivalent to that of a NN 1D Ising model with M links and N_d domain walls. The excess free energy, $\beta \Delta F$, captures the effects of the excluded-volume interactions.

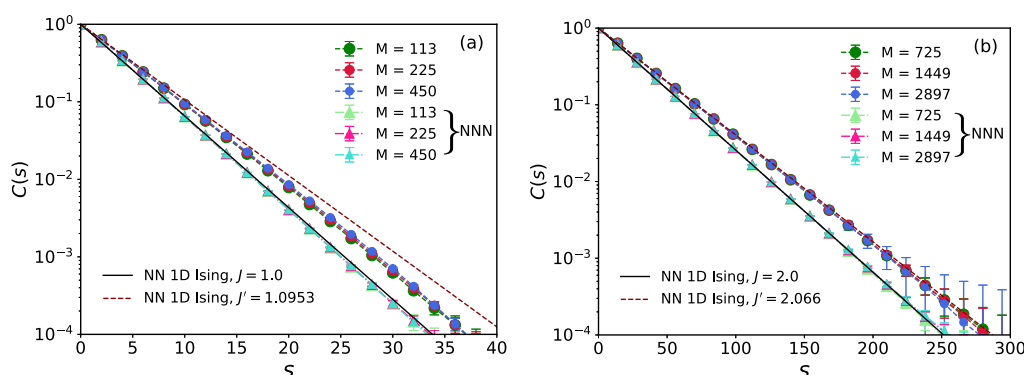


Figure 10. Correlation function of spin variables, $C(s)$, as a function of the contour distance s between two spins. Panels (a) and (b) present $C(s)$ for DHPs with excluded-volume interactions and coupling strengths $J = 1$ and $J = 2$, respectively. For both cases, results are shown for three chain lengths $M = N_k l_k / l_0$ (solid circles), chosen to have the same number of Kuhn segments, N_k , as the three chain lengths considered for $J = 0.5$ in Figure 8. Additionally, solid triangles represent $C(s)$ for the same chain lengths, obtained in simulations with only next-nearest-neighbors (NNN) excluded-volume interactions. Black solid lines show $C(s)$ of the NN 1D Ising model with the same J , whereas red dashed lines correspond to $C(s)$ of the NN 1D Ising model with a renormalized J' (for details see main text).

βF_{id} is given by the sum of the free energy of the Ising degrees of freedom and the free energy of an ideal chain

$$\beta F_{id} = J_0(M - 1) + 2J_0 N_d + \ln \left(\frac{M - 1}{N_d} \right) - N_d \ln 4\pi \quad (18)$$

$$\begin{aligned} \beta F_{id} = & -J_0(M - 1) + 2 \underbrace{\left(J_0 - \frac{\ln 4\pi}{2} \right)}_{=J} N_d \\ & + N_d \left[\ln \left(\frac{N_d}{M - 1} \right) \right] + (M - 1 - N_d) \\ & \left[\ln \left(\frac{M - 1 - N_d}{M - 1} \right) \right] \end{aligned} \quad (19)$$

The sum of the first two terms is the exact energy of a spin configuration with N_d walls, whereas the logarithmic term with the binomial coefficient represents the entropy of distributing N_d domain walls on $M - 1$ bonds. The last term, accounts for the free energy of a freely jointed chain with N_d fully flexible hinges. Minimizing βF_{id} with respect to N_d , we recover eq 9.

The excess free energy, $\beta \Delta F$, follows from eq 11. In Figure 11a–c we report $\beta \Delta F$ (black open circles), normalized by $\langle N_d \rangle$, as a function of $\langle N_d \rangle$ for $J = 0.5$, 1 and 2, respectively. Additionally, the same figure presents the excess energy $\beta \Delta E$ (red open circles) and excess entropy $\Delta S/k_B$ (blue open circles), per domain wall. $\beta \Delta E$ is calculated as the difference between the average energy of spin–spin interactions in configurations of chains with excluded volume and ideal chains. For ideal chains, the average energy is given by $\beta \langle E \rangle = -J(M - 1) \tanh(J)$, which corresponds to the average energy of the NN 1D Ising model.⁵⁶ The excess entropy is then obtained using the relation $\Delta S/k_B = \beta \Delta E - \beta \Delta F$.

As anticipated, the excluded volume favors conformations with lower energy ($\beta \Delta E < 0$) and reduces entropy ($\Delta S/k_B < 0$). The excess free energies are accurately described by the universal functional form for self-avoiding polymers (green solid circles), established by des Cloizeaux.^{69,70} To leading orders in inverse powers of $\langle N_d \rangle$, we obtain for the free-energy difference between the freely jointed chain with and without excluded volume

$$\frac{\beta \Delta F}{\langle N_d \rangle} = f - (\gamma - 1) \frac{\ln[\langle N_d \rangle]}{\langle N_d \rangle} + \frac{c}{\langle N_d \rangle} \quad (20)$$

In principle, γ is a universal critical exponent.⁷¹ However, the chains modeled in our study are rather short, particularly when measured in terms of number of homochiral domains. Therefore, we cannot expect to extract universal coefficients⁴⁴ and treat γ as an empirical fitting parameter, similarly to the constants, f and c . The asymptotic limit of $\beta \Delta F / \langle N_d \rangle \rightarrow f$ for $\langle N_d \rangle \gg 1$, obtained from the fitting procedure, is shown in Figure 11a–c by the green dashed line.

Thus, the first key conclusion is that the total free energy adopts the form

$$\beta F \simeq f N_d - J_0(M - 1) + 2J N_d + \ln \left(\frac{M - 1}{N_d} \right) \quad (21)$$

for $\langle N_d \rangle \gg 1$. This is nothing but the (approximate) free energy of a NN 1D Ising model with an effective, renormalized spin coupling

$$J' = J + \frac{f}{2} \quad (22)$$

supporting our conjecture in Section 4 that for infinitely long chains the spin–spin correlation function will converge to the exponentially decaying one of a NN 1D Ising model with an effective, renormalized coupling. Substituting the fitted values of f , we estimate the effective, renormalized coupling as $J' \simeq 0.6097$, 1.0865, and 2.067 for $J = 0.5$, 1, and 2 respectively. These estimates closely match the values obtained from $J' = \text{atanh}[C(s = 1)]$ (see Figures 8 and 10). The plots for $\beta \Delta E$ and $\Delta S/k_B$ demonstrate that f is determined by a combination of entropic and enthalpic effects.

A second notable observation is that the number of homochiral domains at which $\beta \Delta F / \langle N_d \rangle$ starts to approach the infinite-chain limit f decreases as J becomes larger. For example, for $J = 0.5$, 1, and 2 a relative deviation $(f - \beta \Delta F / \langle N_d \rangle) / f \sim 0.05$ occurs at $\langle N_d \rangle \simeq 22$, 10, and 1.6, respectively. The saturation of the (normalized) $\beta \Delta E$ and $\Delta S/k_B$ follows a similar trend. This observation suggests that for strong coupling $J \gtrsim 2$, the statistics of spin variables (local helicity) is influenced by excluded-volume interactions between parts of the DHP corresponding to only a few sequential homochiral domains. In

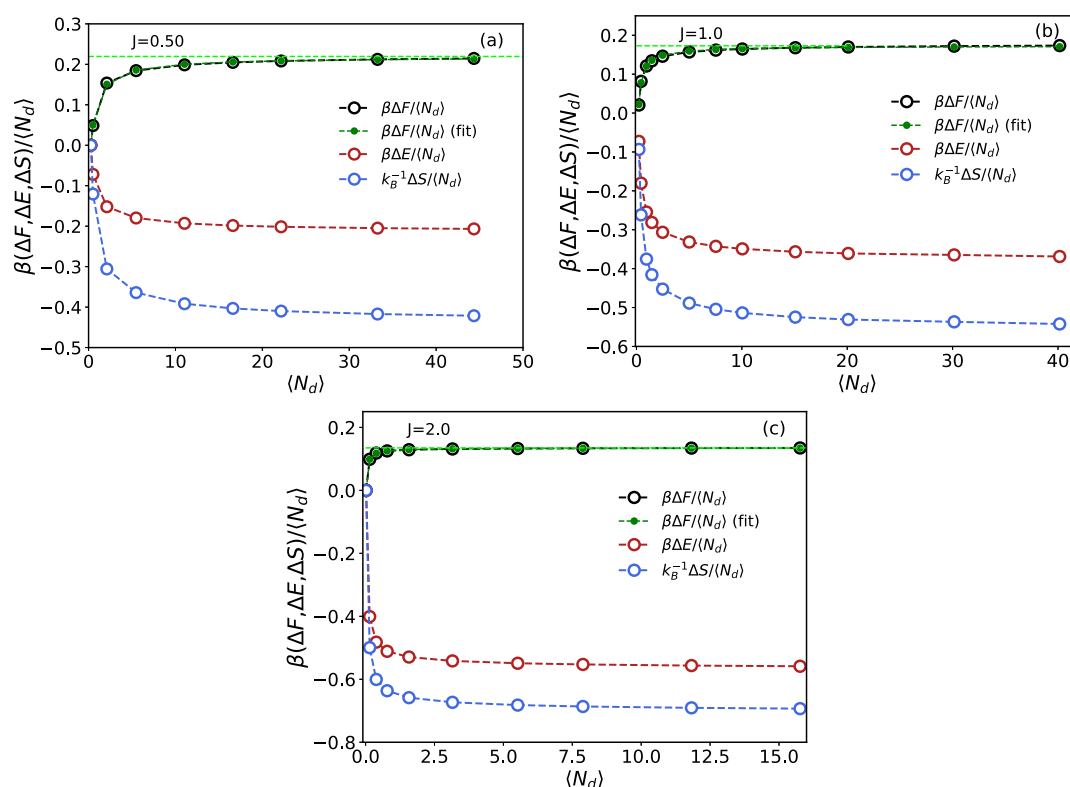


Figure 11. Excess free energy $\beta \Delta F$, excess energy $\beta \Delta E$, and excess entropy $\Delta S/k_B$ as a function of the average number of domain walls $\langle N_d \rangle$. Panels (a–c) correspond to spin–spin coupling parameters $J = 0.5, 1$, and 2 , respectively. Each panel includes the fit of $\beta \Delta F$ to eq 20 (solid green circles) and the horizontal green dashed line indicating the asymptotic limit $\beta \Delta F / \langle N_d \rangle \rightarrow f$ for $\langle N_d \rangle \gg 1$, as determined from the fit.

other words, for large J the effect of excluded volume on the spin degrees of freedom is rather local when contour distances, s , are measured in units of the size of an average homochiral domain.

6. CONCLUSIONS AND OUTLOOK

The properties of cooperative helical order in dynamical helical polymers (DHPs) are commonly explained^{4,17–19} by reducing these polymers to ferromagnetic 1D Ising models and assuming only nearest-neighbor (NN) interactions between spins. In this study, we investigated whether the excluded-volume interactions along the polymer chain can induce an effective non-local coupling between spins, thereby challenging this assumption.

To carry out this investigation, we proposed a generic, minimal model of DHPs that combines the NN 1D Ising description of polymer helicity with an explicit representation of polymer conformations in 3D space. The spin variables, $\sigma_i = \pm 1$ (i is the position of the spin along the chain contour), represent a local right- or left-handed helical sense and the strength of coupling between NN spins is given by the cooperativity parameter, J , of helical order. Our approach draws inspiration from an established model^{38–40} of freely-jointed chains (FJC) with flexible hinges. The model's configurational space comprises both, the spin degrees of freedom and the FJC segment positions, and was sampled via a Monte Carlo scheme.

When excluded-volume interactions between FJC segments are turned off, the statistics of the spin variables (local helical sense) decouples from polymer conformations and follows the behavior of a NN 1D Ising model with coupling strength J . Excluded-volume interactions couple the two “sub-systems” and

create effective non-local interactions between spins. These effective interactions, though present, decay fast enough with the spin distance s along the chain contour, such that the critical behavior remains unaffected. Namely, the helical order parameter (or magnetization in the Ising analog), still belongs to the universality class of NN 1D Ising models.

The analysis of spin–spin pair-correlation function, $C(s)$, and the excess free-energy calculation suggest that, for infinitely long chains, the collective effect of excluded-volume interactions consists in mapping the DHP onto a NN 1D Ising model with an effective, renormalized coupling strength $J' > J$. As a result, the spin–spin correlation length remains finite. This observation demonstrates that the consequences of chain self-avoidance are different in the two “sub-systems”, spin variables and chain conformations. In the case of chain conformations, bond–bond correlation, $C_b(s)$, in long self-avoiding polymers is known to decay algebraically,^{23,24} implying an infinite bond–orientation correlation length.

$C(s)$ of chains with finite length exhibits two characteristic finite-size effects, providing direct evidence of non-local spin–spin coupling: (i) the values of $C(s)$ at fixed s increase as chains become longer, with $C(s)$ asymptotically approaching the exponentially decaying form of $C(s)$ for a NN 1D Ising model with an effective renormalized spin coupling J' . (ii) The crossover of $C(s)$ from exponential to a non-exponential decay shifts to larger s as chains become longer. Interestingly, these finite-size effects decrease as the coupling strength J increases.

To summarize, our results suggest that the parameters J , extracted by mapping DHPs onto NN 1D Ising models, contain long-range contributions from chain self-avoidance. As a result, these parameters may overestimate, to some extent, the free-energy cost associated with helical reversals.

Certainly, this conjecture warrants further investigation. To highlight the effect of excluded-volume interactions, we employed a minimal model that strongly couples 1D spin configurations and 3D polymer conformations. Specifically, chains were only allowed to bend at domain wall between homochiral domains, i.e. at a point of helix reversal. Real DHPs possess an additional mechanism of flexibility,^{14,26,72} which is decoupled from local helical sense because it is based on thermal fluctuations of angular degrees of freedom within homochiral domains. The relative importance of the two flexibility mechanisms depends on the polymer chemistry and external conditions. Long-range effects from excluded-volume interactions are not expected to significantly affect the statistics of the local helical sense in DHPs where the second flexibility mechanism is dominant. One example of such polymers are some poly(*n*-alkyl isocyanates).¹⁴ In contrast, it has been suggested that helical reversals are the dominant flexibility mechanism in polysilylenes at high temperatures.^{25,26} The second mechanism of chain flexibility could be incorporated into our model, and it would be interesting to explore whether it indeed reduces the effect of excluded volume on helicity.

Furthermore, our results suggest that the impact of chain self-avoidance on local helicity decreases as *J* increases. For typical DHPs, such as polysilylenes and poly(*n*-alkyl isocyanates), the cooperativity parameters are large, ranging from *J* = 1.5 to 3.4.^{1,14,15,26,27} (In some cases, the value of the cooperativity parameter reported in the literature must be divided by a factor of 2^{15,17,73} to match the *J* in eq 1). Therefore, one might expect that the renormalization of *J* due to excluded-volume interactions is negligible for these DHPs.

However, studies of cooperative helical order in DHPs often involve chiral bias, with typical examples being the sergeants-and-soldiers⁵ and majority-rule⁶ effects. These phenomena are described by NN 1D Ising models in a random field.^{15,16,19} Depending on the specific conditions, variations in the random field can significantly increase¹⁵ the number of helical reversals, even in DHPs with large *J*. In such cases, the influence of chain self-avoidance on helical order may once again become non-negligible. Finally, novel helical polymers with intrinsically weak cooperativity have also been reported.⁷⁴

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.4c03197>.

Definition of the function $K(N_k)$. Bond–bond correlation functions for *J* = 1. Comparison of l_k and $l_{k,ex}$ for different *J*. Explanation of behavior of spin–spin correlation function for the NNN case (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Kostas Ch Daoulas – Max Planck Institute for Polymer Research, Mainz 55128, Germany; orcid.org/0000-0001-9278-6036; Email: daoulas@mpip-mainz.mpg.de

Authors

Keerti Chauhan – Max Planck Institute for Polymer Research, Mainz 55128, Germany

Marcus Müller – Institute for Theoretical Physics, Georg-August University, Göttingen 37077, Germany; orcid.org/0000-0002-7472-973X

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.macromol.4c03197>

Funding

Open access funded by Max Planck Society.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors express their appreciation to Robinson Cortes-Huerto for carefully reading this manuscript. K.C. and K.D. are grateful to the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) for funding: Project number 233630050-TRR 146. M.M. acknowledges financial support by the DFG under grant MU 1674/15-2.

■ REFERENCES

- (1) Yashima, E.; Maeda, K.; Iida, H.; Furusho, Y.; Nagai, K. Helical Polymers: Synthesis, Structures, and Functions. *Chem. Rev.* **2009**, *109*, 6102–6211.
- (2) Cornelissen, J. J. L. M.; Rowan, A. E.; Nolte, R. J. M.; Sommerdijk, N. A. J. M. Chiral Architectures from Macromolecular Building Blocks. *Chem. Rev.* **2001**, *101*, 4039–4070.
- (3) Mayer, S.; Zentel, R. Chiral Polyisocyanates, a Special Class of Helical Polymers. *Prog. Polym. Sci.* **2001**, *26*, 1973–2013.
- (4) Green, M. M.; Peterson, N. C.; Sato, T.; Teramoto, A.; Cook, R.; Lifson, S. A Helical Polymer with a Cooperative Response to Chiral Information. *Science* **1995**, *268*, 1860–1866.
- (5) Green, M. M.; Reidy, M. P.; Johnson, R. D.; Darling, G.; O’Leary, D. J.; Willson, G. Macromolecular Stereochemistry: the Out-of-Proportion Influence of Optically Active Comonomers on the Conformational Characteristics of Polyisocyanates. The Sergeants and Soldiers Experiment. *J. Am. Chem. Soc.* **1989**, *111*, 6452–6454.
- (6) Green, M.; Garetz, B. A.; Munoz, B.; Chang, H.-P.; Hoke, S.; Cook, R. G. Majority rules in the copolymerization of mirror image isomers. *J. Am. Chem. Soc.* **1995**, *117*, 4181–4182.
- (7) Yashima, E.; Maeda, K. Chirality-Responsive Helical Polymers. *Macromolecules* **2008**, *41*, 3–12.
- (8) Verswyvel, M.; Koeckelberghs, G. Chirality in Conjugated Polymers: When Two Components Meet. *Polym. Chem.* **2012**, *3*, 3203–3216.
- (9) Lago-Silva, M.; Fernández-Míguez, M.; Rodríguez, R.; Quiñoá, E.; Freire, F. Stimuli-responsive Synthetic Helical Polymers. *Chem. Soc. Rev.* **2024**, *53*, 793–852.
- (10) Yashima, E.; Matsushima, T.; Okamoto, Y. Chirality Assignment of Amines and Amino Alcohols Based on Circular Dichroism Induced by Helix Formation of a Stereoregular Poly((4-carboxyphenyl)-acetylene) through Acid-Base Complexation. *J. Am. Chem. Soc.* **1997**, *119*, 6345–6459.
- (11) Lee, D.; Jin, Y.-J.; Kim, H.; Suzuki, N.; Fujiki, M.; Sakaguchi, T.; Kim, S. K.; Lee, W.-E.; Kwak, G. Solvent-to-Polymer Chirality Transfer in Intramolecular Stack Structure. *Macromolecules* **2012**, *45*, 5379–5386.
- (12) Dellaportas, P.; Jones, R. G.; Holder, S. J. Induction of Preferential Helical Screw Senses in Optically Inactive Polysilanes via Chiral Solvation. *Macromol. Rapid Commun.* **2002**, *23*, 99–103.
- (13) Nagata, Y.; Takeda, R.; Sugimoto, M. Asymmetric Catalysis in Chiral Solvents: Chirality Transfer with Amplification of Homochirality through a Helical Macromolecular Scaffold. *ACS Cent. Sci.* **2019**, *5*, 1235–1240.
- (14) Lifson, S.; Andreola, C.; Peterson, N. C.; Green, M. M. Macromolecular Stereochemistry: Helical Sense Preference in Optically Active Polyisocyanates. Amplification of a Conformational Equilibrium Deuterium Isotope Effect. *J. Am. Chem. Soc.* **1989**, *111*, 8850–8858.
- (15) Selinger, J. V.; Selinger, R. L. B. Theory of Chiral Order in random Copolymers. *Phys. Rev. Lett.* **1996**, *76*, 58–61.

- (16) Selinger, J. V.; Selinger, R. L. B. Cooperative Chiral Order in Copolymers of Chiral and Achiral Units. *Phys. Rev. E* **1997**, *55*, 1728–1731.
- (17) Green, M. M.; Park, J.-W.; Sato, T.; Teramoto, A.; Lifson, S.; Selinger, R. L. B.; Selinger, J. V. The Macromolecular Route to Chiral Amplification. *Angew. Chem., Int. Ed.* **1999**, *38*, 3138–3154.
- (18) Teramoto, A. Cooperative Conformational Transitions in Linear Macromolecules Undergoing Chiral Perturbations. *Prog. Polym. Sci.* **2001**, *26*, 667–720.
- (19) Teraoka, I. *Statistical Thermodynamics*; Wiley: Hoboken, USA, 2019.
- (20) Van Gestel, J.; van der Schoot, P.; Michels, M. A. J. Amplification of Chirality in Helical Supramolecular Polymers. *Macromolecules* **2003**, *36*, 6668–6673.
- (21) Gestel, J. v.; van der Schoot, P.; Michels, M. A. J. Amplification of Chirality in Helical Supramolecular Polymers Beyond the Long-Chain Limit. *J. Chem. Phys.* **2004**, *120*, 8253–8261.
- (22) Palmans, A. R. A.; Meijer, E. W. Amplification of Chirality in Dynamic Supramolecular Aggregates. *Angew. Chem., Int. Ed.* **2007**, *46*, 8948–8968.
- (23) Wittmer, J. P.; Meyer, H.; Baschnagel, J.; Johner, A.; Obukhov, S.; Mattioni, L.; Müller, M.; Semenov, A. N. Long Range Bond-Bond Correlations in Dense Polymer Solutions. *Phys. Rev. Lett.* **2004**, *93*, 147801.
- (24) Schäfer, L.; Ostendorf, A.; Hager, J. Scaling of the Correlations among Segment Directions of a Self-Repelling Polymer Chain. *J. Phys. A: Math. Gen.* **1999**, *32*, 7875–7899.
- (25) Sato, T.; Terao, K.; Teramoto, A.; Fujiki, M. Conformational Fluctuations of Helical Poly(dialkyl silylene)s in Solution. *Macromolecules* **2002**, *35*, 2141–2148.
- (26) Sato, T.; Terao, K.; Teramoto, A.; Fujiki, M. Molecular Properties of Helical Polysilylenes in Solution. *Polymer* **2003**, *44*, 5477–5495.
- (27) Young, J. A.; Cook, R. C. Helix Reversal Motion in Polysocyanates. *Macromolecules* **2001**, *34*, 3646–3653.
- (28) Alemán, C.; Green, M. M. A Molecular Orbital Approach to Conformational Study of the Polysocyanates. *Macromol. Theory Simul.* **2001**, *10*, 100–107.
- (29) Hansmann, U. H.; Okamoto, Y. Finite-Size Scaling of Helix-Coil Transitions in Poly-Alanine Studied by Multicanonical Simulations. *J. Chem. Phys.* **1999**, *110*, 1267–1276.
- (30) Boehm, C. R.; Terentjev, E. M. Minimal Model of Intrinsic Chirality to Study the Folding Behavior of Helical Polymers. *Macromolecules* **2014**, *47*, 6086–6094.
- (31) Zimm, B. H.; Bragg, J. K. Theory of the Phase Transition between Helix and Random Coil in Polypeptide Chains. *J. Chem. Phys.* **1959**, *31*, 526–535.
- (32) Lifson, S.; Roig, A. On the Theory of Helix-Coil Transition in Polypeptides. *J. Chem. Phys.* **1961**, *34*, 1963–1974.
- (33) Landau, L. D.; Lifshitz, E. M. *Statistical Physics*; Pergamon: London, 1959; Vol. 5.
- (34) Varshney, V.; Dirama, T. E.; Sen, T. Z.; Carri, G. A. A. Minimal Model for the Helix-Coil Transition of Wormlike Polymers. Insights from Monte Carlo Simulations and Theoretical Implications. *Macromolecules* **2004**, *37*, 8794–8804.
- (35) Peng, Y.; Hansmann, U. H. E.; Alves, N. A. Solution effects and the order of the helix–coil transition in polyalanine. *J. Chem. Phys.* **2003**, *118*, 2374–2380.
- (36) Furukawa, S.; Takeuchi, K.; Shimana, M. Crystal Structures of Poly(di-n-butyl silane) and Poly(di-n-pentyl silane). *J. Phys.: Condens. Matter* **1994**, *6*, 11007–11014.
- (37) Green, M. M.; Zanella, S.; Gu, H.; Sato, T.; Gottarelli, G.; Jha, S. K.; Spada, G. P.; Schoevaars, A. M.; Feringa, B. L.; Teramoto, A. Mechanism of the Transformation of a Stiff Polymer Lyotropic Nematic Liquid Crystal to the Cholesteric State by Dopant-Mediated Chiral Information Transfer. *J. Am. Chem. Soc.* **1998**, *120*, 9810–9817.
- (38) Khokhlov, A. R.; Semenov, A. N. Liquid-Crystalline Ordering in Solutions of Semiflexible Macromolecules with Rotational-Isomeric Flexibility. *Macromolecules* **1984**, *17*, 2678–2685.
- (39) Glatting, G.; Winkler, R. G.; Reineker, P. Freely Jointed Chain with Variable Segment Number and Length. *Colloid Polym. Sci.* **1995**, *273*, 32–37.
- (40) Noh, G.; Benetatos, P. Tensile Elasticity of a Freely Jointed Chain with Reversible Hinges. *Soft Matter* **2021**, *17*, 3333–3345.
- (41) Natsume, T.; Wu, L.; Sato, T.; Terao, K.; Teramoto, A.; Fujiki, M. Chain-Stiffness and Lyotropic Liquid Crystallinity of Polysilylene Bearing (S)-2-Methylbutyl and n-Decyl Substituents. *Macromolecules* **2001**, *34*, 7899–7904.
- (42) Denisov, S. I.; Hänggi, P. Domain Statistics in a Finite Ising Chain. *Phys. Rev. E* **2005**, *71*, 046137.
- (43) Cifra, P.; Romanov, A. On the Entropy of Single Flexible Polymer Chains. *Makromol. Chem.* **1986**, *187*, 2289–2297.
- (44) Kremer, K.; Baumgärtner, A.; Binder, K. Monte Carlo Renormalization of Hard Sphere Polymer Chains in Two to Five Dimensions. *Z. Physik B Con Mat* **1981**, *40*, 331–341.
- (45) Hsu, H.-P.; Paul, W.; Binder, K. Standard Definitions of Persistence Length Do Not Describe the Local "Intrinsic" Stiffness of Real Polymer Chains. *Macromolecules* **2010**, *43*, 3094–3102.
- (46) Hsu, H.-P.; Paul, W.; Binder, K. Estimation of Persistence Lengths of Semiflexible Polymers: Insight from Simulations. *Polym. Sci. Ser. C* **2013**, *55*, 39–59.
- (47) Grosberg, A. Y.; Khokhlov, A. R. *Statistical Physics of Macromolecules*; AIP Press: New York, 1994.
- (48) Rubinstein, M.; Colby, R. H. *Polymer Physics*; Oxford Univ Press, 2003.
- (49) Terao, K.; Terao, Y.; Teramoto, A.; Nakamura, N.; Fujiki, M.; Sato, T. Temperature and Solvent Dependence of Stiffness of Polyn-hexyl-[(S)-3-methylpentyl]silylene in Dilute Solutions. *Macromolecules* **2001**, *34*, 4519–4525.
- (50) Shimada, J.; Yamakawa, H. Statistical Mechanics of Helical Worm-like Chains. XV. Excluded-volume Effects. *J. Chem. Phys.* **1986**, *85*, 591–600.
- (51) Yamakawa, H. *Modern Theory of Polymer Solutions*; Harper & Row: New York, 1971.
- (52) Schäfer, L. *Excluded Vol. Effects in Polymer Solutions*; Springer: Berlin, Heidelberg, 1999.
- (53) Le Guillou, J. C.; Zinn-Justin, J. Critical Exponents for the n-Vector Model in Three Dimensions from Field Theory. *Phys. Rev. Lett.* **1977**, *39*, 95–98.
- (54) Grassberger, P.; Sutter, P.; Schäfer, L. Field Theoretic and Monte Carlo Analysis of the Domb–Joyce Model. *J. Phys. A: Math. Gen.* **1997**, *30*, 7039–7056.
- (55) Antal, T.; Droz, M.; Rácz, Z. Probability Distribution of Magnetization in the One-dimensional Ising Model: Effects of Boundary Conditions. *J. Phys. A: Math. Gen.* **2004**, *37*, 1465–1478.
- (56) Nishimori, H.; Ortiz, G. *Elements of Phase Transitions and Critical Phenomena*; Oxford Univ Press, 2010.
- (57) Kac, M.; Thompson, C. J. Critical Behavior of Several Lattice Models with Long-Range Interaction. *J. Math. Phys.* **1969**, *10*, 1373–1386.
- (58) Dyson, F. Existence of a Phase-Transition in a One-Dimensional Ising Ferromagnet. *Commun. math. Phys.* **1969**, *12*, 91–107.
- (59) Luijten, E.; Blöte, H. Classical Critical Behavior of Spin Models with Long-Range Interactions. *Phys. Rev. B* **1997**, *56*, 8945–8958.
- (60) Kosterlitz, J. M. Phase Transitions in Long-Range Ferromagnetic Chains. *Phys. Rev. Lett.* **1976**, *37*, 1577–1580.
- (61) Bhattacharjee, J.; Chakravarty, S.; Richardson, J. L.; Scalapino, D. J. Some Properties of a One-Dimensional Ising Chain with an Inverse-Square Interaction. *Phys. Rev. B* **1981**, *24*, 3862–3865.
- (62) Kosterlitz, J.; Thouless, D. Ordering, Metastability and Phase Transitions in Two-Dimensional Systems. *J. Phys. C: Solid State Phys.* **1973**, *6*, 1181–1203.
- (63) Ruelle, D. Statistical Mechanics of a One-Dimensional Lattice Gas. *Commun. Math. Phys.* **1968**, *9*, 267–278.
- (64) Corberi, F.; Lippiello, E.; Politi, P. Universality in the Time Correlations of the Long-Range 1D Ising Model. *J. Stat. Mech.: Theory Exp.* **2019**, *2019*, 074002.

- (65) Binder, K. Finite Size Scaling Analysis of Ising Model Block Distribution Functions. *Z. Phys. B* **1981**, *43*, 119–140.
- (66) Binder, K. In *Computational Methods in Field Theory*; Gausterer, H., Lang, C. B., Eds.; Springer Berlin Heidelberg: Berlin, Heidelberg, 1992; pp 59–125.
- (67) Janke, W. In *Computational Many-Particle Physics*; Fehske, H., Schneider, R., Weiße, A., Eds.; Springer Berlin Heidelberg: Berlin, Heidelberg, 2008; pp 79–140.
- (68) Stephenson, J. Two One-Dimensional Ising Models with Disorder Points. *Can. J. Phys.* **1970**, *48*, 1724–1734.
- (69) des Cloizeaux, J. The statistics of long chains with non-Markovian repulsive interactions and the minimal gaussian approximation. *J. Phys. France* **1970**, *31*, 715–736.
- (70) des Cloizeaux, J. On the Absence of Flory Terms in the Energy and in the Entropy of a Polymer Chain. *J. Physique* **1976**, *37*, 431–434.
- (71) des Cloizeaux, J. Lagrangian Theory of a Self-Avoiding Random Chain. *Phys. Rev. A* **1974**, *10*, 1665–1669.
- (72) Gu, H.; Nakamura, Y.; Sato, T.; Teramoto, A.; Green, M. M.; Andreola, C. Global Conformations of Chiral Polyisocyanates in Dilute Solution. *Polymer* **1999**, *40*, 849–856.
- (73) Jha, S.; Cheon, K.-S.; Green, M. M.; Selinger, J. V. Chiral Optical Properties of a Helical Polymer Synthesized from Nearly Racemic Chiral Monomers Highly Diluted with Achiral Monomers. *J. Am. Chem. Soc.* **1999**, *121*, 1665–1673.
- (74) Rosales, A. M.; Murnen, H. K.; Kline, S. R.; Zuckermann, R. N.; Segalman, R. A. Determination of the Persistence Length of helical and Non-Helical Polypeptoids in Solution. *Soft Matter* **2012**, *8*, 3673–3680.



CAS BIOFINDER DISCOVERY PLATFORM™

STOP DIGGING THROUGH DATA —START MAKING DISCOVERIES

CAS BioFinder helps you find the
right biological insights in seconds

Start your search



A Division of the
American Chemical Society